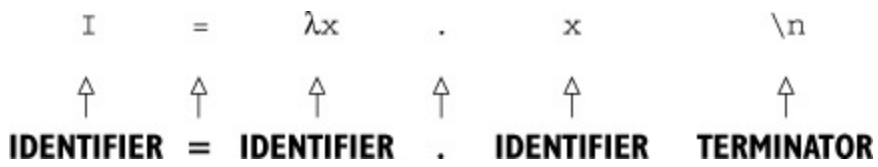
8.4.2. Rewriting the token stream

A slightly less quick-and-dirty way to metaprogram (compared to evaluating strings) is to directly modify the token stream before the AST is generated. By doing that, you're adding your own rewriter to the compilation process. Remember Scruffy's desired syntax?

```
I = λx.x
```

How is it tokenized now?



Your rewriter will need to see the λ symbol and know that it needs to rewrite the rest of the expression to a function:

```
coffee = require 'coffee-script'

scruffyCoffee = '''
I = λx.x
'''

tokens = coffee.compile scruffyCoffee

i = 0
while token = tokens[i]
  # handle token
  i++
```

If the token is a λ, then rewrite expression to a function.

In the next listing you can see a new implementation of ScruffyCoffee that uses a custom rewriter instead of a nasty-looking regular expression.

Listing 8.7. Custom rewriter